

Hongrui (Sam) Sheng

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EDUCATION

Fudan University (FDU)

Bachelor of Science in Chemistry, GPA: 3.85/4.00 | Rank: 9/99

· Key Coursework: Quantum Chemistry (Honor): A+

Shanghai, China

09/2023 - Expected 06/2027

Stanford University

International Honors Program Participant, GPA: 4.11/4.30

· Key Coursework: Introduction to Probability Theory: A+; Programming Abstractions: A

Stanford, CA, USA

06/2025 - 08/2025

PUBLICATIONS

1. P. Walther[†], **H. Sheng**[†], X. Liu[†], B. Feng, R. Coyle, X. Yan, K. Smith, H. Kayal, S. C. Pal, and Z. Zheng*, "Predicting Scale-Up of Metal-Organic Framework Syntheses with Large Language Models," *arXiv preprint*, 2026. <https://arxiv.org/abs/2604.20899>
2. C. Lei, Y. Meng, **H. Sheng**, S. Wang*, X. Wang, and M. Gong*, "The Interplay of Ions and Surfaces during Deposition/Dissolution Equilibrium Enables Efficient Neutral-pH Biomass Electrooxidation," *in submission*, 2026.
3. Z. Li[†], **H. Sheng**[†], Z. Zheng, "Imperfection-Aware Active Learning for Experiment-Grounded Scientific Simulators," *Sim2Science Workshop (Tiny Paper Track), NeurIPS 2026*, in submission, 2026.

[†] Equal contribution; * Corresponding author.

RESEARCH EXPERIENCE

Predictive Synthesis of Metal-Organic Frameworks (MOFs)

St. Louis, MO, USA

Visiting Researcher | Professor Zhiling (Zach) Zheng, Washington University in St. Louis

04/2026 - present

· ActMOF: A 6-Million-Condition Benchmark for Active Learning in MOF Syntheses

- Benchmarked 39 active learning policies (GP, RL, ML, LLM) across a 6M-condition MOF synthesis space as an open-source, reproducible pipeline; GP policies reached top-1% crystallinity within a 120-experiment budget.
- Conducted UMAP analysis, revealing that high-crystallinity conditions cluster into sparse, localized regions.
- Validated framework transferability experimentally by synthesizing and characterizing six MOFs via PXRD and BET, confirming strong agreement with predictions.

· ScaleMOF: Predicting Scale-up of MOF Syntheses

- Redesigned the PU-learning split ($P_s/P_a/U$ for training) to recover unused negative examples for gold-standard evaluation; led the resubmission underlying the model's 93.5% balanced accuracy and 97.7% ROC-AUC.
- Re-benchmarked the fine-tuned LLM against Logistic Regression/Random Forest baselines on 1,536-dim text embeddings; validated dataset diversity via t-SNE.

· ReinforceCOF: Reinforcement-Learning-Guided Generative Design of Covalent Organic Frameworks

- Built a GRPO-based reinforcement learning pipeline for *de novo* COF generation, coupling pyCOFBuilder with SmVAE/RFcode representations to navigate high-dimensional reticular chemical space; diagnosed and reported an upstream generation bug in pyCOFBuilder via GitHub issue.
- Formulated a multi-objective reward function balancing synthesizability, structural novelty, and target-property alignment via SCScore, SAScore, and Tanimoto similarity metrics.
- Engineered a budget-constrained, GCE-based cloud experimentation pipeline supporting large-scale RL training runs.

AI-Guided Bimetallic Electrocatalysts for Glycerol-to-Lactic Acid Conversion

Shanghai, China

Independent Research Leader, Wangdao Research Program | Professor Ming Gong, FDU

05/2025 - present

- Curated literature datasets of 52 entries and engineered β function-inspired biasing factors to encode complex chemical interactions for downstream model training.

- Benchmarked ML models coupled with heuristic optimization (GA, PSO) to screen 1 million metal alloy combinations, identifying candidates with a predicted lactic acid selectivity of $\sim 70\%$.
- Validated top computational candidates through wet-lab synthesis, then further optimized reaction conditions by evaluating electrolyte effects (NaOH) and pulse electrolysis, boosting lactic acid selectivity by $\sim 7\%$.
- Achieved a peak experimental lactic acid selectivity of 56.55% with electrocatalyst $\text{Pt}_{66}\text{Cu}_{34}$.

Biomass Valorization via Electrocatalysis

Shanghai, China

Research Assistant | Professor Ming Gong, FDU

03/2024 - 04/2025

- Synthesized catalysts and conducted electrochemical testing for the targeted electrooxidation of 5-hydroxymethylfurfural (HMF) to 2,5-furandicarboxylic acid (FDCA) under neutral conditions.
- Investigated the catalytic role of Co^{2+} and quantified product yields using UV-Vis spectroscopy and HPLC, applying foundational laboratory techniques to troubleshoot downstream product isolation and purification challenges.

SELECTED PROJECTS

AI-Driven Molecular Modeling and Simulation | [Codes on GitHub](#)

Shanghai, China

AI in Computational Chemistry and Simulation, Course Project

03/2026 - 06/2026

- Applied equivariant neural networks (PaiNN) and diffusion models (GeoDiff) using PyTorch and e3nn to predict atomic forces and generate 3D molecular conformations.
- Programmed custom Nosé-Hoover NVT integrators and Stochastic Surface Walking (SSW) algorithms via Python and LASP to explore complex potential energy surfaces and thermodynamic transitions.
- Fine-tuned Chemformer for single-step retrosynthesis, deploying an [interactive web application](#), and applied unsupervised clustering (PCA, t-SNE) on QM9 datasets to map molecular properties.

Automated Computational Workflow for S_N2 Mechanisms | [Codes on GitHub](#)

Shanghai, China

Computational Chemistry, Course Project

11/2025 - 12/2025

- Automated Python-Gaussian workflows for 7 different amine methylation, engineering self-correcting protocols to resolve optimization failures in sterically hindered outliers.
- Applied SISSO-inspired regression ($R^2 > 0.95$) linking extracted electronic (Q_N) and topological (κ_3) descriptors to activation energies.

SKILLS

Programming	Python, C++, $\text{\LaTeX}$
Computational Chemistry	Psi4, Gaussian, MACE
Miscellaneous Tools	Claude Code, Grok Bot; Google Cloud; CrystalMaker
Languages	Chinese (Native), English (TOEFL iBT 107/120), Japanese (JLPT N1)

PROFESSIONAL ACTIVITIES & AFFILIATIONS

Peer Reviewer (Co-reviewer): *Nature Communications*, 2026

Member: American Chemical Society (ACS), American Institute of Chemical Engineers (AIChE), 2026

LEADERSHIP

Fudan Integrative Medicine Association & Traditional Chinese Medicine Club

Shanghai, China

Founding/Executive President

03/2024 - 09/2025

- Established the first student organization promoting interdisciplinary medicine at Fudan.
- Designed automated outreach workflows using Python, expanding membership to 100+ students.

HONORS & AWARDS

Outstanding Student, Fudan University, 2025

Second Prize, Syensqo Scholarship, Syensqo (formerly Solvay), 2025

Second Prize, Scholarship for Outstanding Students, Fudan University, 2024, 2025